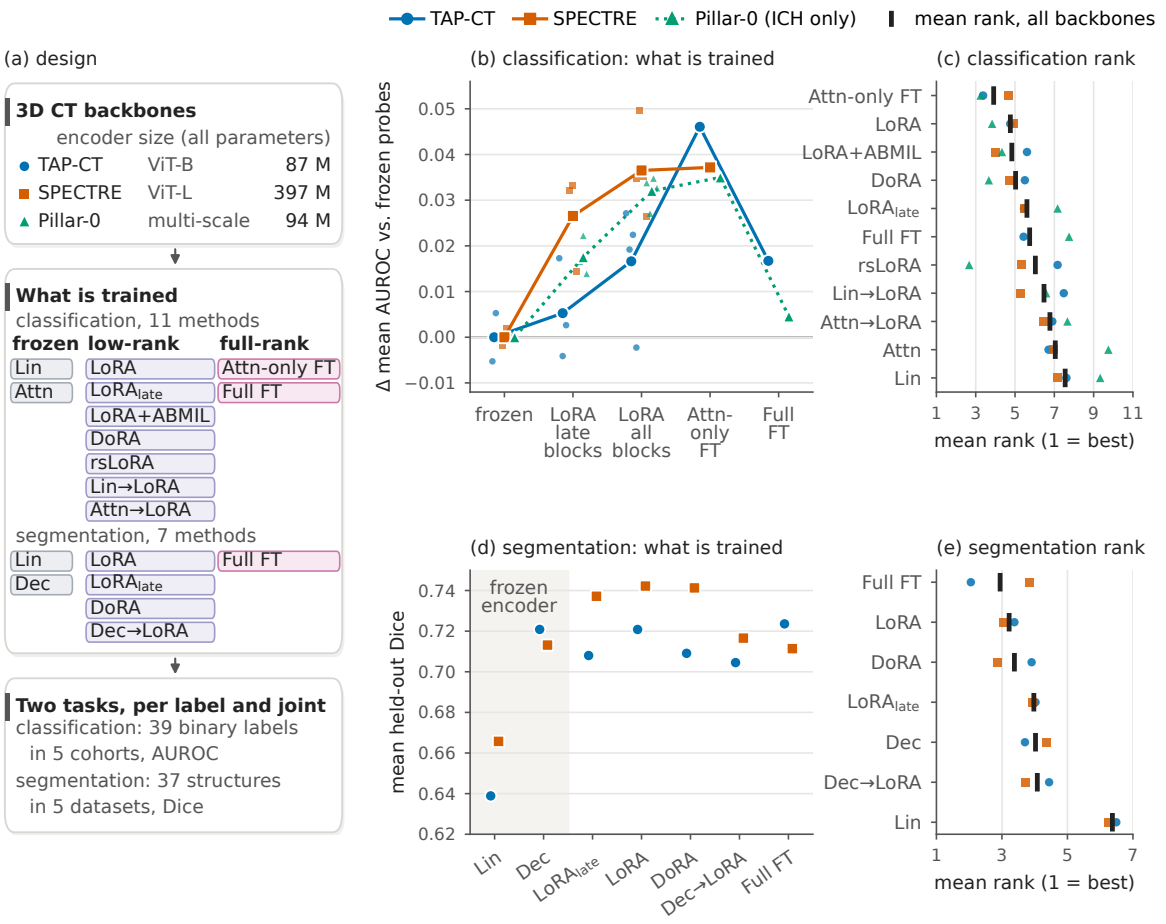
TAP-CT
SPECTRE
Pillar-0 (ICH only)
mean rank, all backbones
(a) design
3D CT backbones
encoder size (all parameters)
TAP-CT ViT-B 87 M
SPECTRE ViT-L 397 M
Pillar-0 multi-scale 94 M
What is trained
classification, 11 methods
frozen
low-rank
full-rank
Lin
Attn
LoRA
LoRA_late
LoRA+ABMIL
DoRA
rsLoRA
Lin→LoRA
Attn→LoRA
Attn-only FT
Full FT
segmentation, 7 methods
Lin
Dec
LoRA
LoRA_late
DoRA
Dec→LoRA
Full FT
Two tasks, per label and joint
classification: 39 binary labels
in 5 cohorts, AUROC
segmentation: 37 structures
in 5 datasets, Dice
(b) classification: what is trained
Δ mean AUROC vs. frozen probes
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
frozen
LoRA late blocks
LoRA all blocks
Attn-only FT
Full FT
(c) classification rank
Attn-only FT
LoRA
LoRA+ABMIL
DoRA
LoRA_late
Full FT
rsLoRA
Lin→LoRA
Attn→LoRA
Attn
Lin
1 3 5 7 9 11
mean rank (1 = best)
(d) segmentation: what is trained
mean held-out Dice
0.74
0.72
0.70
0.68
0.66
0.64
0.62
frozen encoder
Lin
Dec
LoRA_late
LoRA
DoRA
Dec→LoRA
Full FT
(e) segmentation rank
Full FT
LoRA
DoRA
LoRA_late
Dec
Dec→LoRA
Lin
1 3 5 7
mean rank (1 = best)